

#

Module 1

1 star patterns

#code

1 square

```
*****
*****
*****
*****
*****
```

Logic → using two nested loops.

→ outer loop controls row (5R)

→ inner loop print 5 stars in each row.

→ move to next line after every row.

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

2 hollow square pattern

```
*****
*      *
*      *
*      *
*      *
*****
```

Logic → use two nested loops.

② print * when

→ first row

→ last row.

→ first column

→ last column.

→ print space for remaining positions.

```
#include <iostream>
using namespace std;
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n)
                cout << "*";
            else
                cout << " ";
        }
        cout << endl;
    }
    return 0;
}
```


#3 Triangle pattern

```
*
* *
* * *
* * * *
* * * * *
```

- # Logic → outer loop controls row from 1 to 5
 → inner loop prints star equal to row number.
 → No. of stars increase by one in every row.

code

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

#4 Inverted Triangle

```
* * * * *
* * * *
* * *
* * *
```

- # Logic → outer loop starts from 5 and decrease to 1
 → inner loop prints stars equal to current row value.
 → Number of stars decreases by one in every row.

code

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 5; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```


#5 pyramid pattern

```

      *
     ***
    *****
   *****
  *****
 *****

```

Logic → use three loops.

- first loop prints leading space.
- second loop prints odd number of stars ($2*i-1$)
- space decrease and stars increase in every row.

code

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n=5;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i; j++) {
            cout << " ";
        }
        for (int j=1; j<=2*i-1; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}

```

#6 Diamond pattern

```

      *
     ***
    *****
   *****
  *****
 *****

```

Logic → Divide the pattern into two parts.

- ① upper pyramid
- ② lower "

code

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n=5;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i; j++) {
            cout << " ";
        }
        for (int j=1; j<=2*i-1; j++) {
            cout << "*";
        }
        cout << endl;
    }
    for (int i=n-1; i>=1; i--) {
        for (int j=1; j<=n-i; j++) {
            cout << " ";
        }
        for (int j=1; j<=2*i-1; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}

```


⑦ Half Diamond pattern

```
*
**
***
****
*****
****
***
**
*
```

logic

→ first part prints
increase triangle.

② second part prints decreasing triangle.

③ join both part.

④ peak row is printed only
once

⑧ Half Diamond inverted

```
* * * * *
* * * *
* * *
* *
*
* *
* * *
* * * *
* * * * *
```

Logic → ① first print inverted triangle

② Then print normal triangle.

③ number of stars first decrease then increase.

```
# code
#include <iostream>
using namespace std;
int main() {
```

```
for (int i = 1; i <= 5; i++)
```

```
{ for (int j = 1; j <= i; j++)
```

```
cout << " ";
```

```
cout << endl;
```

```
}
```

```
for (int i = 4; i >= 1; i--)
```

```
{ for (int j = 1; j <= i; j++)
```

```
cout << " ";
```

```
cout << endl;
```

```
}
```

```
return 0;
```

```
# include <iostream>
using namespace std;
int main() {
```

```
for (int i = 5; i >= 1; i--)
```

```
{ for (int j = 1; j <= i; j++)
```

```
cout << " ";
```

```
cout << endl;
```

```
}
```

```
for (int i = 2; i <= 5; i++)
```

```
{ for (int j = 1; j <= i; j++)
```

```
cout << " ";
```

```
cout << endl;
```

```
}
```

```
return 0;
```


⑨ Number square

```

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1

```

Logic → ① use two nested loops.

② outer loop prints rows.

③ inner " " prints number 1 repeatedly.

```

#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cout << "1";
        }
        cout << endl;
    }
    return 0;
}

```

⑩ Incrementing Number pattern

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

```

Logic → outer loop controls rows.

② inner loop prints most from 1 to 5.

③ same sequence is repeated in every row.

```

#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}

```


⑪ Right Triangle Number pattern

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
  
```

Logic → ① outer loop control rows
 ② inner loop prints numbers from 1 up to current row number
 ③ number increase in each row.

```

#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
  
```

⑫ Inverted Diamond Number pattern

```

2 2 2 2 3
4 4 4 4
5 5 5
6 6
7
6 6
5 5 5
4 4 4 4
3 3 3 3 3
  
```

Logic → ① Divide pattern into two parts.
 ② First half → number increase
 → Repetitions (3, 4, 5, 6, 7)
 ③ 2nd half → number decreases
 (6, 5, 4, 3)
 → Repetitions increase.
 Together they form an inverted diamond shape.

```

#include <iostream>
using namespace std;

int main() {
    int num = 2;
    for (int i = 5; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << num << " ";
        }
        cout << endl;
        num++;
    }
    num = 6;
    for (int i = 2; i <= 5; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << num << " ";
        }
        cout << endl;
        num--;
    }
    return 0;
}
  
```























